# Vernacular AI E-Commerce Enabler

Business & Impact Plan

**Document 2:** Implementation Roadmap & Socio-Economic Impact Analysis

**Participant:** Tanmay Dey | **Institution:** Brainware University

## 1. The Market Gap: Status Quo vs. Proposed Solution

Currently, traditional weavers in the Shantipur and Phulia clusters rely heavily on wholesale middlemen because the digital barrier to entry for direct-to-consumer sales is too high.

| Feature | Current E-Commerce (Amazon/Flipkart) | Vernacular AI Enabler (Our Solution) |
|---|---|---|
| **Literacy Requirement** | High (English/Hindi typing required) | Zero (Voice-based input in Bengali) |
| **Product Photography** | Requires studio lighting / white background | Processes standard smartphone photos |
| **SEO & Tagging** | Requires digital marketing knowledge | AI auto-generates tags and categories |
| **Time to List** | 15–30 minutes per item | < 2 minutes per item |
| **Target User** | Urban Sellers / Aggregators | Rural Grassroots Artisans |

## 2. Socio-Economic Impact

- **Margin Improvement:** By sharing WhatsApp catalogs directly with urban retail buyers or boutique owners, artisans bypass the 30–40% margin typically taken by regional wholesalers.
- **Digital Inclusion:** Empowers women weavers and elderly artisans who have smartphone access but lack English typing skills, bringing them into the digital economy.
- **Cultural Preservation:** By making it profitable to sell authentic Taant and Jamdani sarees directly, the platform incentivizes the continuation of traditional weaving arts over transitioning to power-looms.

## 3. Pilot Phase Budget Estimate (₹20,000)

To launch a functional pilot in the Nadia district, the following bootstrapped budget is required:

| Item | Amount | What it covers |
|---|---|---|
| **Cloud Infrastructure** | **₹5,000** | Vercel hosting for Next.js frontend, Python backend hosting, and PostgreSQL database deployment for 3 months. |
| **AI API Credits** | **₹5,000** | OpenRouter/Gemini API credits for processing image and audio data during the pilot phase (estimated 2,000 product generations). |
| **Field Operations & Onboarding** | **₹10,000** | Travel to Phulia, printed local-language onboarding materials, and small incentives for the first 20 beta-testers. |
| **Total** | **₹20,000** | Pilot phase, Nadia district |

## 4. Scalability Vector

While Phase 1 targets Bengali-speaking weavers, the platform scales across the Eastern Region by simply altering the AI prompt to process Hindi, Bhojpuri, or Maithili voice notes. The cloud architecture (Next.js + Python) requires zero modification to serve artisans in Bihar or Jharkhand.